

João Jerónimo

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🔗 JoaoJeronimoHW

Summary

Applied Scientist and PhD Economist with 4+ years modeling how prices, incentives, and supply-and-demand interact in large-scale markets. Specialised in elasticity and behavioural-response modeling, causal inference (A/B testing, DiD, IV, Synthetic Control), and decision systems under uncertainty and delayed feedback. Proven track record turning problem formulation into production-grade pipelines and counterfactual simulations of agent behaviour under changing incentives – directly applicable to dynamic pricing and two-sided marketplace optimization.

Work experience

AI Evaluation Expert

Remote

micro1

Feb 2026 – Present

- Designed domain-specific evaluation frameworks to assess LLM performance on economics and data science tasks, including rubric development for causal inference workflows, econometric method selection, and code quality for OpenAI.
- Designed synthetic datasets with controlled statistical properties (endogeneity, heteroskedasticity, panel structure) to stress-test estimator behaviour and validate experiment design under realistic, adversarial conditions.
- Authored expert-level task specifications simulating senior economist instruction, used to evaluate model capability across regression, forecasting, and causal identification tasks.

Data Scientist (Doctoral Researcher)

Edinburgh, UK

Heriot-Watt University

Sep 2021 – Nov 2025

- Modeled price (wage) elasticities of supply on a 50M+ row two-sided panel, showing they move from 2.7 to 0.6 across the cycle; diagnosed a lagging, noisy feedback signal producing sign-flipped estimates and replaced it with a contemporaneous flow signal to restore stable, interpretable results.
- Cut macro forecast error by $\approx 60\%$ (RMSE) by benchmarking an LSTM pipeline against the incumbent ARIMA model on a UK government forecasting dataset.
- Built a queryable graph of worker flows across 357K firms — a two-sided matching network — enabling causal inference on sparse subsets where standard estimators break down.
- Built a counterfactual simulation engine in Python to forecast agent responses to price and incentive changes, replacing a static policy model with a behavioural, response-based one.
- Communicated complex modelling results to non-specialist audiences, including conference presentations and undergraduate instruction.

Data Research Analyst

Lisbon, Portugal

University of Lisbon

Jan 2019 – Jun 2021

- Pre-processing and streamlining data across multiple formats (.csv, .json and HTML)
- Automation of data extraction processes through web scraping using Python, resulting in data preparation time reduction from 5-6 days to under 2 hours.
- Writing VBA scripts for simultaneous operations in large matrices, significantly speeding up data operations required for pre-processing.

Skills

Pricing & marketplace modeling: elasticity & supply/demand response, structural/dynamic optimization, incentive & behavioural modeling, two-sided matching.

Causal inference & experimentation: A/B testing, DiD (incl. staggered adoption), IV/2SLS (weak-instrument diagnostics), Synthetic Control, RDD, Double/Debiased ML (DML).

Modeling: forecasting & time series (ARIMA, LSTM), Bayesian econometrics, high-dimensional/regularised regression (LASSO).

Engineering: Python (Advanced), SQL, R (Advanced), DuckDB, Git/GitHub; production-style data pipelines, reproducible workflows.

Independent projects

Iberian Electricity Market Data Pipeline

github.com/mibel 

- Built a production-style pipeline from scratch to ingest, clean, and model hourly market observations (OMIE, ENTSO-E, Open-Meteo) into a DuckDB warehouse and an analysis-ready panel – to evaluate the Iberian gas-price cap, a real market-design / price-intervention question.
- Tools Used: Python (pandas, requests, entsoe-py), DuckDB, SQL, Open-Meteo API, Git/GitHub, Jupyter

Macroeconomic Forecasting Engine

github.io.com/sfc 

- Built LSTM pipeline, benchmarked against ARIMA to enhance forecasting accuracy of a UK government forecasting model by close to 60% (in terms of RMSE).
- Tools Used: Python (scikit-learn, TensorFlow)

Medium-term Inflation Predictor

github.com/ml 

- Wrote and calibrated ARIMA, VAR and Bayesian UC-SV models to predict quarterly inflation. Pseudo out-of-sample model evaluation, robust standard errors and comparative forecast table.
- Tools Used: Python (numpy, SciPy, pandas)

IV Bias Correction Algorithm

github.com/macro 

- Implemented a bias-correction estimator in R/Stata to solve endogeneity issues in time-series data. Based on Wang and Bellemare (2019).
- Tools Used: R, Stata

Education

Heriot-Watt University

Oct 2021 – Jan 2026

PhD in Economics

- Research areas: macroeconomics, labour economics, applied econometrics, data science. Award: UK Research and Innovation (UKRI) Scholarship (≈£60,000).

University of Minho

Sept 2017 – Jan 2020

MSc in Economics (summa cum laude)

- Honors: National Award for Academic Merit (Top ranked students in Portugal, 2019); "Best Student" scholarships for highest GPA in 2018.

University of Minho

Sept 2014 – Jun 2017

BA in International Relations (magna cum laude)

- Honors: "Best Student" scholarships for highest GPA in 2016.

Journal publications

Interactions between financial constraints and economic growth

Jul 2023

J. Jerónimo, A. Azevedo, P.C. Neves, M. Thompson

The North American Journal of Economics and Finance, vol. 67

[10.1016/j.najef.2023.101943](https://doi.org/10.1016/j.najef.2023.101943) 